

Combiner.R

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Combine_inventories	<i>Create gridded methane emissions maps that include all sectors</i>
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Description

'Combine_inventories' writes a netcdf for every unique combination of sectoral emissions estimates. Optionally writes additional netcdf with the thermogenic and nonthermogenic sectors separately.

Usage

```
Combine_inventories(output_directory, separate_thermo)
```

Arguments

`output_directory`
Character providing the full filepath to save processed data

`separate_thermo`
Logical. Pulled from config file. Indicating whether or not to save partial inventories: one that is only thermogenic sources, and one that is only non-thermogenic sources.

Details

This function considers every possible output across all sectors and filters to keep those that have been saved to the output folder. From there it creates a table of every possible unique combination of sectors. If many options have been enabled, this can be over 1,000 combinations, so caution is warranted. Additionally, if the config option is set, three inventories will be created for each unique combination instead:

- One that is the total across all sectors
- One that is the total across only thermogenic sources
 - stationary combustion of wood

- natural gas distribution
- natural gas transmission
- Gridded EPA mobile combustion
- Gridded EPA abandoned coal
- Gridded EPA surface coal
- Gridded EPA underground coal
- Gridded EPA petroleum systems exploration
- Gridded EPA petroleum systems production
- Gridded EPA petroleum systems refining
- Gridded EPA petroleum systems transport
- Gridded EPA abandoned oil gas
- Gridded EPA natural gas exploration
- Gridded EPA natural gas processing
- Gridded EPA natural gas production
- Gridded EPA industry petrochemical
- Gridded EPA industry ferroalloy
- One that is the total across only non-thermogenic sources
 - landfills
 - stationary combustion of fossil fuels
 - wastewater
 - wetlands
 - Gridded EPA composting
 - Gridded EPA enteric fermentation
 - Gridded EPA manure management
 - Gridded EPA rice cultivation
 - Gridded EPA field burning

Value

Nothing is returned from the function, but the main outputs are many netcdf files of the methane emissions across all sectors. Given the large number of possible files and variations, they are titled "Combined_inventory_combination_1.nc" increasing numerically. A csv is also saved that details what variations were used for each sector for each inventory.

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Examples

```
library(terra)
Combine_inventories <- function(output_directory=~/.//Desktop/out/",
                                separate_thermo=T)
```

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